Problem Statement No.14: AI Agent for Smart Farming Advice

The Challenge

Farmers often struggle with low crop yield, unpredictable weather, pest infestations, and limited access to timely expert advice. Traditional farming advisory services are fragmented, slow, and not personalized to the farmer's specific land, crop, or climate conditions. With the rise of climate change, data-driven smart farming solutions are needed to empower farmers with actionable insights.

The Objective

Develop an AI-powered Smart Farming Agent that provides personalized, real-time, and sustainable agricultural advice. The solution should:

1. **Agricultural Knowledge Retrieval (RAG) –** Use Retrieval-Augmented Generation to fetch best practices, crop-specific guidelines, and government advisory updates.
2. **Weather & Soil Insights** – Integrate real-time weather forecasts, soil health data, and irrigation needs to optimize farm planning.
3. **Pest & Disease Detection –** Enable multimodal support where farmers upload crop images for early pest/disease identification and get instant recommendations.
4. **Personalized Advisory Agent –** Suggest seeds, fertilizers, irrigation schedules, and harvesting timelines based on farmer profile and land conditions.
5. **Market & Cost Optimization –** Forecast demand, suggest market prices, and recommend cost-efficient practices for better income.
6. **Multi-Agent Collaboration –**
    - Crop Advisory Agent
    - Weather & Irrigation Agent
    - Pest/Disease Detection Agent
    - Market Insights Agent

Tech Stack (Mandatory)

Use **any one** of the following approaches:

- LangFlow + IBM watsonx.ai + Granite Models
- IBM Cloud Agentic Lab with IBM Granite Models
- Replicate with LangChain + RAG + IBM Granite Models